UNIT II · CHEAT SHEET

Aerodynamics, airframes and UAV design

Sizing, lift and drag, Reynolds number, drag polar, configurations, regulations, stealth, control surfaces

Design starts with the mission

- Mission and operational requirements → ConOps → initial mass W_0 from mass fractions → payload and mass budget.
- Bound the design with wing loading W/S and power loading P/W (or T/W), then run trade studies.

Core equations

- **Lift:** $L = \frac{1}{2}\rho V^2 S \cdot C_L$
- **Drag:** $D = \frac{1}{2}\rho V^2 S \cdot C_D$
- **Reynolds:** $Re = \rho V \bar{c} / \mu$ (inertial ÷ viscous)
- **Drag polar:** $C_D = C_{D0} + K \cdot C_L^2$, $K = 1/(\pi \cdot AR \cdot e)$
- **Best L/D:** at $C_{Di} = C_{D0}$, $C_{L,opt} = \sqrt{(C_{D0}/K)}$, $(L/D)_{max} = 1/(2\sqrt{(C_{D0} \cdot K)})$
- **Stall:** $V_{stall} = \sqrt{(2W/(\rho S \cdot C_{L,max}))}$
- **Wing loading:** $W/S = m_0 g/S$ (N/m²)
- **Cruise power:** $P = D \cdot V = (W/(L/D)) \cdot V$
- **Tip Mach:** $M_{tip} = \pi \cdot n \cdot D/a$
- **Hinge moment:** $\tau = q \cdot S_e \cdot c_e \cdot C_h \cdot \delta \cdot FOS$

Low-Reynolds-number flight

- Small UAV wings fly at $10^4 < Re < 5 \times 10^5$.
- Laminar separation bubbles raise drag; choose low-Re airfoils (Selig, Drela families).

Which airframe?

- **Fixed-wing:** best range and endurance; needs runway, launcher or net.
- **Multirotor:** hover anywhere, simple mechanics; high induced power, short endurance.
- **Single-rotor helicopter:** efficient hover, heavy lift; complex mechanics.
- **Hybrid VTOL:** no runway plus fixed-wing cruise; extra mass and transition risk.

Regulations at a glance

- **USA (FAA):** Part 107; operations-over-people categories 1–4.
- **EU (EASA):** Open / Specific (SORA, SAIL) / Certified.
- **UK (CAA):** CAP 722 guidance.
- **India (DGCA):** Drone Rules 2021: nano ≤ 250 g, micro 250 g–2 kg, small 2–25 kg, medium 25–150 kg, large > 150 kg; green, yellow, red zones; type certification.

Stealth and signatures

- **Acoustic:** keep $M_{tip} < 0.45$; more blades, lower rpm.
- **Visual:** counter-shading, slender planform.
- **Thermal:** shield exhaust, cooling plenums.
- **Radar:** edge alignment, faceting, radar-absorbent material.

Control surfaces

- Aileron → roll (p); elevator → pitch (q); rudder → yaw (r).
- Elevons (roll + pitch) and ruddervators (pitch + yaw) combine axes.
- Size servos for hinge moment at maximum dive dynamic pressure, with a factor of safety.

Exam traps

- Using V instead of V^2 in lift or drag.
- Forgetting that best L/D needs parasite = induced drag.
- Specs without a verification matrix (requirement → method → pass/fail).